

TYPED CONTROL, INFORMATION, AND SADDLE ENVELOPES FOR KINETIC COTANGENT PHASES

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ABSTRACT. Weak convergence of joint laws does not make disintegration continuous. We therefore keep a global weak belief state but prove Feller filtering on model-derived dominated Sobolev charts that control unnormalized kernels and their observation derivatives. Atomic microscopic observations enter by an exponentially good regularization. The unnormalized positive kernel, its evidence, and its projective zero-evidence direction form an exact Markov belief state with measurable dynamic programming. Tolerance-dependent bounded-Lipschitz coordinates approximate the full value, and the exact regular experiment is QMD, LAN, and locally Bernstein–von Mises.

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1. ROUND-SEVENTEEN POSITIVE CLOSURE: DOMINATED FILTER CHARTS, EXACT BELIEF DYNAMICS, AND QUADRATIC-MEAN STATISTICS

Disintegration is not continuous for weak convergence of arbitrary joint laws. Accordingly, the global belief state carries the weak topology, while Feller continuity and local asymptotic statistics are proved on a model-derived

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regular chart whose topology directly controls unnormalized observation kernels and their derivatives. Singular and zero-evidence strata remain in the global state but are not silently imported into the regular chart.

1.1. The global belief state and regular observation charts. Let E be the microscopic or path state and Y the observation space. A belief is a probability $\pi \in \mathcal{P}_W(E)$ with finite Lyapunov moment. For a control a let

$$\Lambda_\pi^a(dx, dy) = \pi(dx)K^a(x, dy)$$

be the unnormalized one-step joint kernel. The exact next belief is a Rokhlin conditional law of x' given the observed y under the predicted joint law. The global state stores the unnormalized coefficient measure and, when its total mass is zero, a projective direction in the positive cone.

Fix a reference observation measure ν . A regular chart $\mathfrak{B}_{M,c,C,s}$ consists of beliefs and controls for which the predicted observation kernel has a density $k_\pi^a(x, y)$ satisfying

$$c \leq k_\pi^a(x, y) \leq C, \quad \|k_\pi^a\|_{W_y^{s,1}(C_{W,x}^1)} \leq M, \quad s > \dim Y.$$

The chart metric is the weighted weak belief metric plus the $W_y^{s,1}(L_W^1)$ distance of unnormalized kernels. By Rellich compactness, its bounded subsets are compact in the corresponding L^1 kernel topology.

Theorem 1.1 (Model-derived regular filter chart). *For the finite-cylinder Sinai observations and the regular hard-sphere particle/contact observations used in this series, every compact source and control set has a chart $\mathfrak{B}_{M,c,C,s}$ on a fixed positive-evidence region. The microscopic joint kernels enter that chart after an exponentially good smoothing at a scale tending to zero, and the smoothing error is exponentially negligible at the governing LDP speed.*

Proof. For Sinai observations, the unsmoothed vector–return–roof coefficient theorem of A2 gives a density with uniform derivatives on compact interior observation windows. Convolution only regularizes the bounded local path coordinates and its error is controlled by the singularity shield of A3. For hard spheres, B1’s exact-number coefficient theorem and B2’s positive actual-contact source expansion give the same bounds on compact regular windows. In both cases the denominator has a uniform Gaussian-scale lower bound, while Sobolev derivatives follow by differentiating the common source kernel before inversion. The complement of the regular event has an arbitrarily large exponential cost after the chart is enlarged. Atomic finite-volume empirical observations are therefore represented by an exponentially good regularized kernel, not asserted to lie literally in an L^1 density topology. $\square$

1.2. Integrated filter stability. If $J(dx, dy) = j(x, y)m(dx)\nu(dy)$ is in the chart, put $e_J(y) = \int j(x, y)m(dx)$ and

$$\Pi_J(y)(dx) = \frac{j(x, y)m(dx)}{e_J(y)}.$$

Theorem 1.2 (Dominated disintegration estimate). *For two joint kernels $J, \tilde{J}$ in one regular chart,*

$$\int W_1^W(\Pi_J(y), \Pi_{\tilde{J}}(y)) \bar{e}(y) \nu(dy) \leq C_{\mathfrak{B}} \|j - \tilde{j}\|_{L_y^1(L_W^1)},$$

where $\bar{e} = e_J \wedge e_{\tilde{J}}$. Consequently prediction followed by observation is Feller on the regular chart. The theorem makes no continuity claim for arbitrary weakly convergent joint laws.

Proof. For every weighted Lipschitz test φ ,

$$\left| \frac{\int \varphi j dm}{e_J} - \frac{\int \varphi \tilde{j} d\tilde{m}}{e_{\tilde{J}}} \right| \leq \frac{C}{c} \int W(x) |j(x, y) - \tilde{j}(x, y)| dm_0(x),$$

after embedding both kernels in the fixed dominating measure m_0 of the chart. Take the supremum over the weighted Lipschitz unit ball and integrate in y . The positive lower bound on evidence and the fixed dominating measure are precisely the hypotheses absent from weak disintegration. The prediction kernel is continuous in the chart norm by Theorem 1.1; composition proves Feller continuity. $\square$

1.3. Exact belief Markov state and dynamic programming. Let $\mathcal{Z}$ be the disjoint union of the regular belief charts and the blown-up zero/singular-evidence boundary. The state records the unnormalized positive coefficient measure, its mass, and a projective cone direction when the mass is zero.

Theorem 1.3 (Belief transition and posterior DPP). *For each predictable control pair there is a Borel transition kernel on $\mathcal{Z}$ obtained by prediction and Rokhlin disintegration. On regular charts it is Feller; on the full state it is universally measurable. Compact relaxed controls admit measurable selectors, and the finite-horizon risk-sensitive value satisfies the exact posterior dynamic-programming principle. One-time preparation, reward-only feedback, and adaptive likelihood control remain three distinct games.*

Proof. The microscopic transition and observation kernels are Borel on standard Borel spaces. Rokhlin disintegration can be selected jointly measurably in the joint law on a countably generated observation sigma-field. Adjoin the unnormalized mass and the projective zero-mass direction before normalization; this makes the update Borel also at zero evidence. The regular-chart Feller property is Theorem 1.2. The compact relaxed-control space and lower semicontinuity of the transition action give a measurable minimum/maximum selector by the measurable maximum theorem. Conditioning the exact path likelihood at the next observation time and applying the tower property yields the DPP. Since the phase/preparation variable is fixed once in the first game but the adaptive source is selected blockwise in the third, their timing is not conflated. $\square$

1.4. Finite determining coordinates without a fictitious rate. Let $K \subset \mathcal{P}_W(E)$ be a compact regular belief set. For every $\eta > 0$, choose a finite family $\Phi_\eta = (\phi_1, \dots, \phi_{m(\eta)})$ from the weighted bounded-Lipschitz unit ball that separates an η -net of K . Let R_η map the coordinate vector to a nearest point of the net.

Proposition 1.4 (Tolerance-dependent belief reconstruction). *The families may be chosen measurably so that*

$$\sup_{\pi \in K} W_1^W(\pi, R_\eta(\pi(\Phi_\eta))) \leq 3\eta.$$

The finite-coordinate controlled values converge uniformly to the full belief value as $\eta \downarrow 0$. No claim is made that the first K members of an arbitrary dense enumeration have a prescribed resolution rate.

Proof. Compactness gives a finite η -net. For each pair of distinct net points, Kantorovich duality supplies a bounded Lipschitz test separating them by at least their distance minus η ; collect finitely many such tests. Nearest-net-point reconstruction is Borel after fixing a deterministic tie rule and gives the metric estimate. The regular filter kernel and reward are uniformly continuous on K , so backward induction and the contraction of the risk-sensitive Bellman operator give uniform convergence of values. $\square$

1.5. Quadratic-mean differentiability and LAN. Let $\vartheta \mapsto p_\vartheta^\varepsilon$ be the exact finite-volume density of the observed experiment on a regular chart, with respect to the fixed dominating observation measure. The source derivatives come from the exact finite-volume B1/B2 or A2 coefficient before taking the limit.

Theorem 1.5 (Exact-experiment QMD and LAN). *On every compact identifiable regular chart,*

$$\int \left(\sqrt{p_{\vartheta+h}^\varepsilon} - \sqrt{p_\vartheta^\varepsilon} - \frac{1}{2} h^T \dot{\ell}_{\vartheta,\varepsilon} \sqrt{p_\vartheta^\varepsilon} \right)^2 d\nu = o(|h|^2)$$

uniformly for $|h| \rightarrow 0$, and the score covariance converges to a positive information matrix I_ϑ . Therefore, for $h/\sqrt{\mu_\varepsilon}$,

$$\log \frac{dP_{\vartheta+h/\sqrt{\mu_\varepsilon}}^\varepsilon}{dP_\vartheta^\varepsilon} = h^T \Delta_\varepsilon - \frac{1}{2} h^T I_\vartheta h + o_{P_\vartheta}(1),$$

with $\Delta_\varepsilon \Rightarrow N(0, I_\vartheta)$. A prior with positive continuous density then satisfies a Bernstein–von Mises theorem in the local chart, uniformly over the declared compact strategy class.

Proof. The exact coefficient theorems provide two source derivatives of the density in L^1 and a third derivative dominated in L^1 on the fixed chart. Taylor’s theorem for the square root, using the uniform lower and upper density bounds, gives QMD. The score is the centered derivative of the exact finite-volume log normalizer; A2 or B3 gives its Gaussian limit and B1 supplies the prepared Schur complement. QMD plus the central limit theorem

yields LAN with exact finite-volume centering. Uniform tests follow from compact identifiability and the coefficient large-deviation bound; Laplace expansion of numerator and denominator gives BvM. All upstream inputs invoked here are proved earlier in the dependency order. $\square$

Theorem 1.6 (Round-seventeen C1 closure). *The exact belief process is Markov on an unnormalized positive-kernel state, regular filter charts have model-derived compactness and Feller stability, finite coordinates approximate the full value at a chosen tolerance, and the exact regular experiment is QMD, LAN, and locally Bernstein–von Mises.*

Proof. Combine Theorems 1.1, 1.2, 1.3, and 1.5, together with Proposition 1.4. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Seventeen dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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